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A palaeomagnetic study of the Umvimeela Dyke, Zimbabwe: evidence for a Mesoproterozoic overprint

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Abstract

Palaeomagnetic and magnetic susceptibility results are reported from sixteen new sites (complementing the previous seven) on the Umvimeela Dyke, one of the two major satellite dykes parallelling the Great Dyke of Zimbabwe. Four of the new sites are in the deformed northernmost section of the dyke with a further four just to their south. The magnetisation of the long cratonic section of the Umvimeela Dyke is characterised by a component P, carried principally by magnetite and considered primary, with a mean direction $D=219.8°$, $I=-58.7°$ with $k=116.8$ and $\alpha_{95}=3.3°$. This direction is almost identical to those of the Great Dyke and two other associated intrusions. The angular dispersion of site mean directions for the Umvimeela Dyke is only 6.5°, suggesting a short time of emplacement and cooling.

A second component S, carried by magnetite, titanomagnetite and ilmenohaematite, is present at most sites but in bigger proportions as the Limpopo and Zambezi Belts are approached. Considerations such as geographical distribution, magnetic susceptibility and dyke geology suggest S is an overprint. Its direction ($D\approx 195°$, $I\approx 10°$) is similar to that of the ~1100 Ma Umkondo dolerites and it is suggested that it was introduced chemically at or just before the onset of Pan-African activity in the region.

Of the four sites in the deformed section of the dyke one carries the P component, but the other three have been completely remagnetized. At two of these a component Z was detected which may have been introduced ~650 Ma ago. This suggests that this section, and the accompanying Musengezi Complex of the Great Dyke, were deformed in the late Pan-African orogenic phase of the Zambezi Belt.

1. Introduction

The Great Dyke crosses the Zimbabwe craton in a NNE-trending direction for 550 km. Its surface geometry is misleading; it is not a true dyke at all but the remains of a series of near-contiguous layered igneous bodies emplaced 2461 ± 16 Ma ago (Hamilton, 1977) along a line of weakness in the Earth's crust. Igneous bodies associated with the Great Dyke include the Southern Satellite Dykes beyond the south end of the Great Dyke and within the Limpopo Belt, the Umvimeela and East Dykes roughly parallelling the Great Dyke, gabbroic dykes intruding some of the NNE-trending major faults in the craton (Wilson et al., 1987), and, almost certainly, a sill at Nenga and a dyke at Ngundu Halt, both within the northern zone of the Limpopo Belt. The Great Dyke and its principal satellites are described and analysed by Wilson and Prendergast (1988).

The Umvimeela Dyke (Fig. 1), up to 100 m wide, is situated to the west of the Great Dyke at

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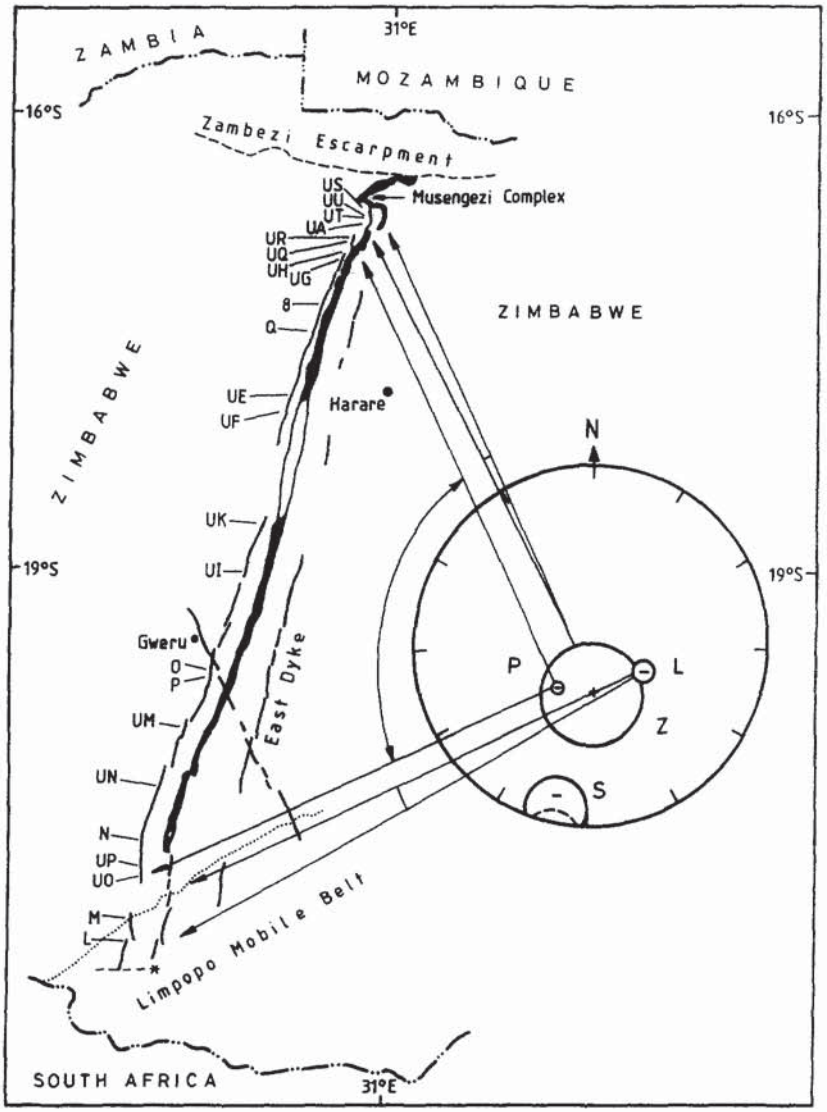


Fig. 1. Sketch map of Zimbabwe showing location of sampling sites from the Umvimeela Dyke. The stereonet shows the components of remanence (L, P, S and Z) isolated in various segments of the dyke. Distances in Table 1 are measured from the line with asterisk.

a distance varying from 1 km in the north to 18 km in the south (Worst, 1960). It continues some 100 km further south than the Great Dyke, crossing into the Limpopo Belt. For most of its length the Umvimeela Dyke is undeformed except for small offsets associated with cross-cutting faults. At its northern end, however, within the Zambezi Belt, the Umvimeela Dyke imitates the southern section of the extensively deformed Musengezi Complex of the Great Dyke; it is concave towards the west and wholly displaced about 10 km to the east.

The Umvimeela Dyke is composed of quartz gabbro. The deformed section of the dyke is highly faulted and there is evidence of extensive shearing at dyke margins. There are marked variations in the degree of alteration of the rock in the deformed section; in some places only

slight alteration is evident, but in others there has been extensive uralitization and saussuritization (Wiles, 1972).

McElhinny and Gough (1963), in their palaeomagnetic study of the Great Dyke, included results from a single site (MG8) in the *undeformed* northern section of the Umvimeela Dyke. The mean direction was not significantly different from that of the Great Dyke. Later Jones et al. (1975) reported results from six sites in the Umvimeela Dyke including two (L and M) from its extension into the Limpopo Belt. They concluded that sites L and M had been completely remagnetized during a thermal event some 1900 Ma ago (Van Breemen and Dodson, 1972). The mean direction of sites deeper into the craton (including MG8) was $N=5$, $D=220.6°$, $I=-62.7°$ with $k=128.6$. The angular dispersion of directions was only 7.1°, smaller than the angular dispersion expected from secular variations of the present Earth's field, suggesting that the Umvimeela Dyke was emplaced over a very short period of time.

The present study extends that of Jones et al. (1975) and includes sites close to and within the deformed zone. It was hoped that a detailed study of the very long undeformed section would provide firmer information on the duration of emplacement and cooling, while sites in the deformed section might provide information on the time (or times) of deformation of the Musengezi Complex (and Zambezi Belt).

2. Sampling

Outcrop, particularly in the northern half of the Umvimeela Dyke, is often difficult to locate. Nevertheless sixteen new sites were collected including four sites (UA, US, UT and UU) within the deformed zone and four sites just to the south of the deformed zone (Fig. 1). At site UG a transect of sampling was conducted consisting of three samples from the fractured edge of the dyke, one sample 16 m away from the edge and four samples from the undeformed centre of the dyke. Two sites, UO and UP, are located slightly north of where the Umvimeela Dyke enters the Limpopo Belt. All samples were hand-drilled and oriented by sun compass. Between five and nine samples were cored at each site.

3. Measurements

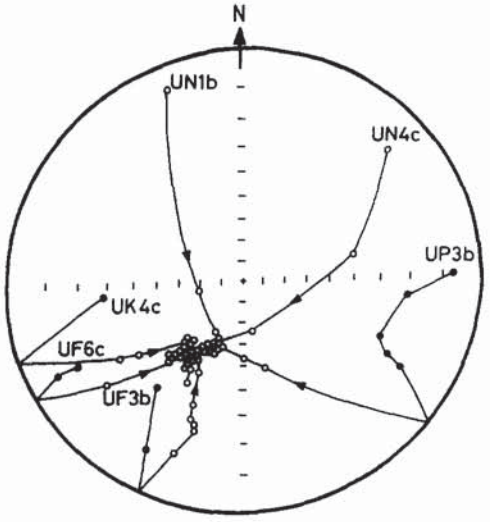
Measurements of intensity and direction of magnetization of specimens were made using a PAR-SM1 magnetometer. Alternating field demagnetization (AF) was performed using apparatus similar to that of McElhinny (1966). One specimen per sample was demagnetized stepwise, usually up to 180 mT; to try to eliminate components of magnetization introduced by the tumbling method, above 30 mT double demagnetization was performed (Hillhouse, 1977). A second specimen per sample was used in thermal demagnetization experiments using a Magnetic Measurements MMTD1 apparatus. The initial magnetic susceptibility of one specimen per sample has been measured using a Bison 3101 apparatus; corrections have been made for both length and diameter of specimens.

Measurements have been analysed using the battery of techniques now available. These include the use of orthogonal vector plots and LINEFIND (Kent et al., 1983), demagnetization circles (Jones et al., 1975; McFadden and McElhinny, 1988), stereonet plots and Fisher (1953) statistics.

4. Results

Five components of magnetization have been identified within Umvimeela Dyke samples. First, the natural remanent magnetization (NRM) includes an isothermal (IRM) component, labelled R. Second, there is a component, labelled P, sometimes persisting up to 180 mT, that is not significantly different from the previously reported direction for the Great Dyke and Umvimeela Dyke. Third, at most sites, there is a component, called S, residing in high-coercivity grains which manifests itself by a southerly shallowing from the P direction at high demagnetizing fields. A further component, Z, is observed

at two sites (UT and UU) in the deformed zone and is believed to be related to Zambezi Belt activity. Finally there is the component L, identified by Jones et al. (1975) at the two sites within the Limpopo Belt. The very different directions



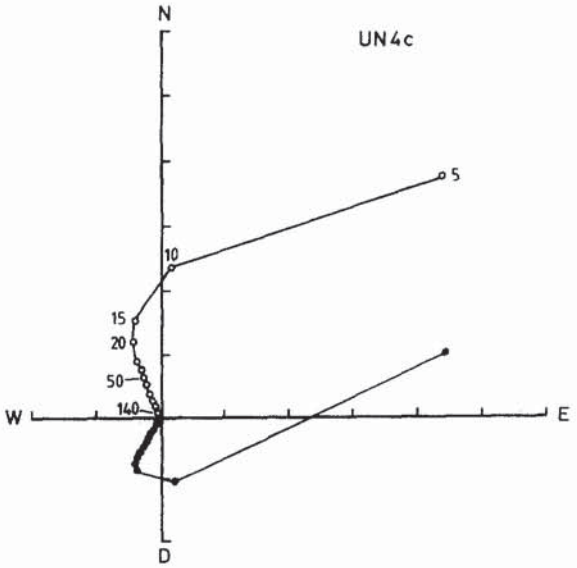


Fig. 2. Equal angle stereonet showing directions of magnetization during AF demagnetization for several specimens from the Umvimeela Dyke. The circles represent negative inclinations. The orthogonal plot shows the removal of the R component during AF and isolation of the P component (axis ticks: ×0.2 A/m). Dots indicate the horizontal component.

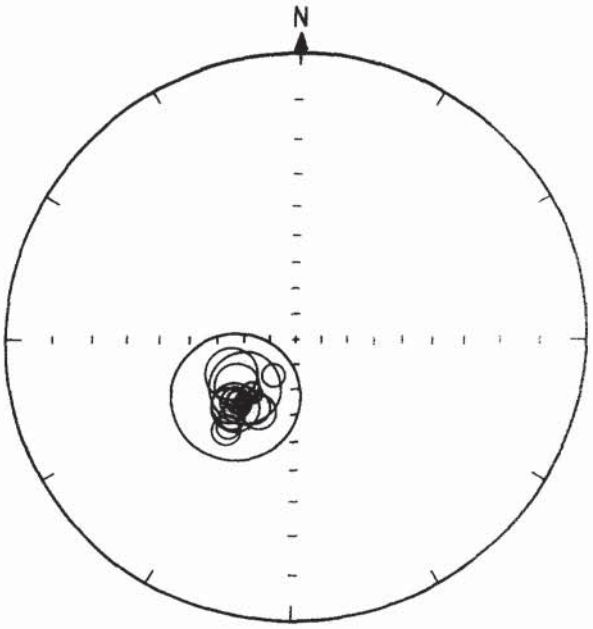


Fig. 3. Equal angle stereonet showing circles of 95% confidence for the site mean directions of component P from the Umvimeela Dyke. All directions have negative inclinations.

associated with components P, S, Z and L are shown in Fig. 1.

4.1. Component R

This component is randomly directed and was the largest component in most specimens. It was usually removed by AF in fields of less than 20 mT while thermal demagnetization of specimens with a significant R component maintained their NRM directions to temperatures over 580°C before intensities dropped to noise level. The low coercivities and high blocking temperatures associated with R suggest that it is carried by multidomain magnetite (Bol'shakov and Shcherbakova, 1979). An IRM origin for R is suggested by its randomly directed nature and comparison of AF decay curves with those of artificially introduced IRMs in the same specimens (Mushayandebvu, 1991).

4.2. Component P

This component is present at all sites except most of those close to or within the Limpopo and

Table 1
Site mean directions (P component), virtual geomagnetic poles (VGP) and magnetic susceptibilities for the Umvimeela Dyke

Site	Lat. (°S)	Long. (°E)	Distance (km)	N	P component				Suscept. ($\times 10^{-3}$)	VGP	
					D (deg.)	I (deg.)	k	α_{95} (deg.)		lat. (°S)	long. (°W)
L*	21.47	29.40	21	6					4.3±0.1		
M*	21.33	29.43	36	6					10.9±0.7		
UO	21.03	29.48	67	6	241.1	−62.3	48.2	9.9	5.5±0.5	4.5	111.0
UP	20.98	29.48	74	5	224.8	−63.5	42.7	13.5	8.6±1.3	12.4	119.9
N*	20.88	29.48	85	4	230.5	−61.5	117	8.5	10.8±1.6	11.5	115.0
UN	20.48	29.58	129	5	210.0	−59.2	167	5.9	4.6±0.9	23.4	125.8
UM	20.11	29.70	170	5/4	224.9	−59.3	21.7	23.3	9.0±0.5	16.7	116.0
P*	19.83	29.87	200	4	210.0	−73.5	385	4.7	9.0±0.4	7.3	135.2
O*	19.70	29.90	215	4	219.0	−59.0	901	3.1	10.7±0.5	20.5	118.9
UI	18.95	30.15	297	5	218.0	−51.7	308	4.4	18.6±2.3	27.1	114.1
UK	18.74	30.19	321	5	224.4	−54.4	138	6.7	8.3±0.6	21.8	117.0
UF	17.95	30.38	407	7	216.2	−48.3	222	4.1	10.5±1.0	31.2	112.6
UE	17.81	30.43	423	9/6	217.4	−62.8	263	4.4	7.4±0.3	19.2	122.1
Q*	17.48	30.53	459	4	222.0	−57.0	260	5.7	5.4±0.3	22.4	114.6
8	17.28	30.60	481	10	217.0	−62.0	474	2.0		20.0	121.5
UG	17.08	30.67	503	8 (3)	221.5	−57.6	177	4.2	15.6±1.6 (1.8±0.4)	22.4	115.1
UH	16.97	30.73	515	5	215.4	−46.5	500	4.1	5.3±0.2	33.6	111.3
UQ	16.88	30.78	526	5	205.3	−58.4	148	6.3	5.3±0.5	29.2	126.9
UR	16.86	30.78	527	9	219.8	−55.6	72.7	8.9	5.8±0.2	25.0	114.4
UA	16.76	30.87	539	5					0.70±0.02		
UT	16.73	30.88	542	5					0.76±0.04		
UU	16.71	30.89	544	5					0.66±0.06		
US	16.67	30.88	549	5	203.0	−41.2	210	5.3	0.79±0.04	43.8	119.4

Notes. Sites marked with * are those of Jones et al. (1975), while site 8 is from McElhinny and Gough (1963). Susceptibility values are given with the standard error; the bracketed entry for site UG is for the edge of the dyke. Distances are measured from the south end of the dyke. N = (number of samples obtained) / (samples used in statistics).

Zambezi Belts, and it is also present at the northern-most site (US). AF demagnetization directions converged towards the P direction during removal of R followed generally by linear segments on orthogonal plots (Fig. 2). In some cases the direction of P was obtained using demagnetization circle analysis. Site mean directions for the P component are listed in Table 1 and shown in Fig. 3. Thermal demagnetization was carried out on specimens after removal of most of R by AF. However, the high blocking temperatures associated with the fraction of R remaining, overlapped with those of P and this led to poorer definition of P using thermal treatment.

P is the dominant component in grains with coercivities ranging from 30 to 180 mT at almost all the sites. The blocking temperature spectra for P show that pure or nearly pure magnetite is its main carrier. The very high coercivities suggest P is carried mainly by single-domain magnetite. However, a number of specimens maintained the P direction during thermal demagnetization up to 640°C after the intensity had dropped sharply just below 580°C, suggesting that ilmenohaematite is a (minor) carrier of the P component too.

4.3. Component S

Many samples at our sites (except UM on the cratonic section where component P itself was poorly defined, and UA, UT and UU in the deformed zone) show a systematic southerly shallowing away from the P direction during demag-

netization. This must be indicative of a further component of NRM which we call S. The shallowing is particularly evident in demagnetizing fields over 100 mT (Figs. 4 and 5). Insights about the origin of S are gained from its within-site variability at site UG as well as its between-site variability. At UG, the three samples collected from the sheared edge of the dyke show the presence of S at 60 mT, whereas at the centre of the dyke P continues to dominate until 100 mT. S is progressively more noticeable as the (Pan African) Zambezi Belt is approached through sites UG, UH, UR and US. At UQ and UR, just south of the deformed zone, the presence of S is clear in fields less than 60 mT, while at US, which has no significant R component, sample NRMs contain P and S in comparable amounts. S becomes slightly more evident as the Limpopo Belt is ap-

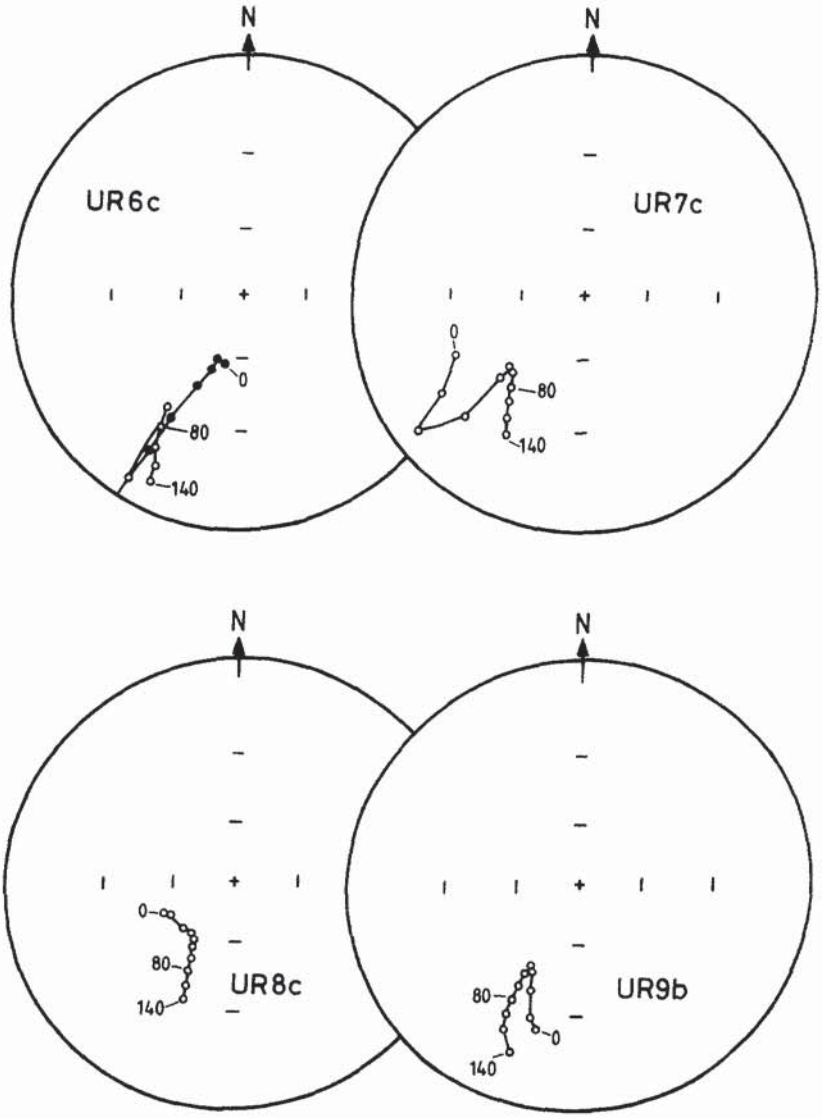


Fig. 4. Equal angle stereonets showing AF trajectories of Umvimeela Dyke specimens. After the initial removal of component R, the southerly shallowing away from component P towards S is evident. Circles represent negative inclinations.

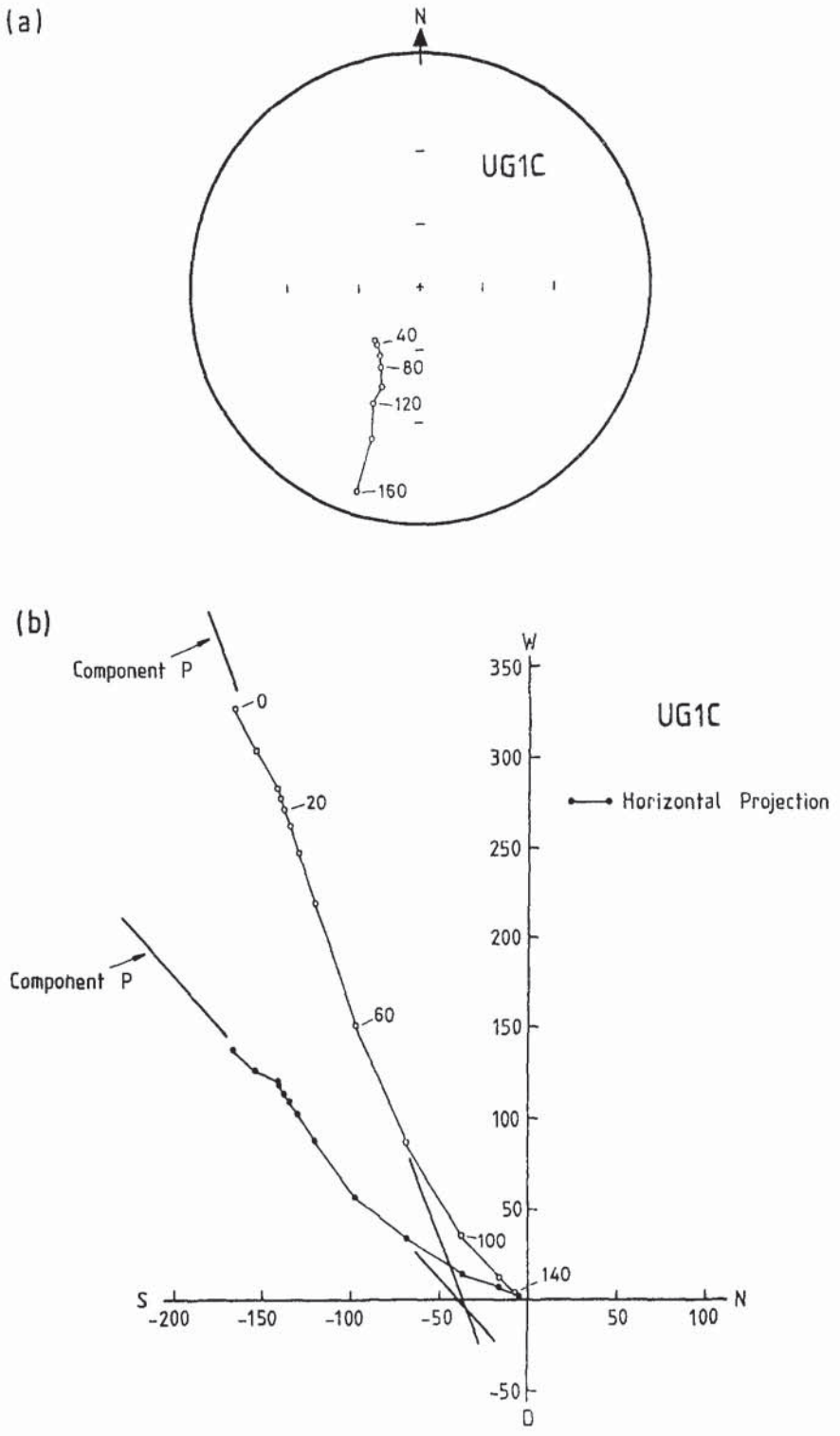


Fig. 5. (a) Equal angle stereonet showing the southerly shallowing away from component P towards S during AF of an Umvimeela Dyke specimen. All directions have negative inclinations. (b) The orthogonal plot shows the P component and the simultaneous removal of P and S between 80 and 140 mT (axis ticks: $\times 10^{-7}$ A m^2).

proached at sites UP and UO (the direction of S is very different, 72°, to that of component L observed by Jones et al., 1975, at sites L and M within the belt).

Specimens whose directions had swung towards S during AF were heated to determine the blocking temperature spectrum of S. Most of these specimens maintained their directions in temperatures above 580°C, showing that the blocking temperature spectra of S and P overlap and that the carriers are the same. However, a number of specimens maintained their direction only up to 550°C and then swung northward back to the P direction which they maintained to over 600°C (Fig. 6). It appears that S is carried by titanomagnetite grains with blocking temperatures not exceeding 560°C. A few specimens showed a slight swing in direction from P to S between 585° and 600°C, suggesting part of S is carried by ilmenohaematite.

Two estimates can be made of the direction of S. Firstly, using demagnetization circle analysis of specimens showing clear movement from P to S, nineteen specimens gave a mean result of $D=198.1°$, $I=-17.1°$ with $k=19.0$ and $\alpha_{95}=8.2°$ (Fig. 7). Secondly, the component removed in the northward swing from S towards P during thermal demagnetization can be estimated by use of LINEFIND and ten specimens gave $D=190.9°$, $I=-6.3°$ with $k=18.4$ and $\alpha_{95}=10.4°$ as an estimate of the S direction using this method.

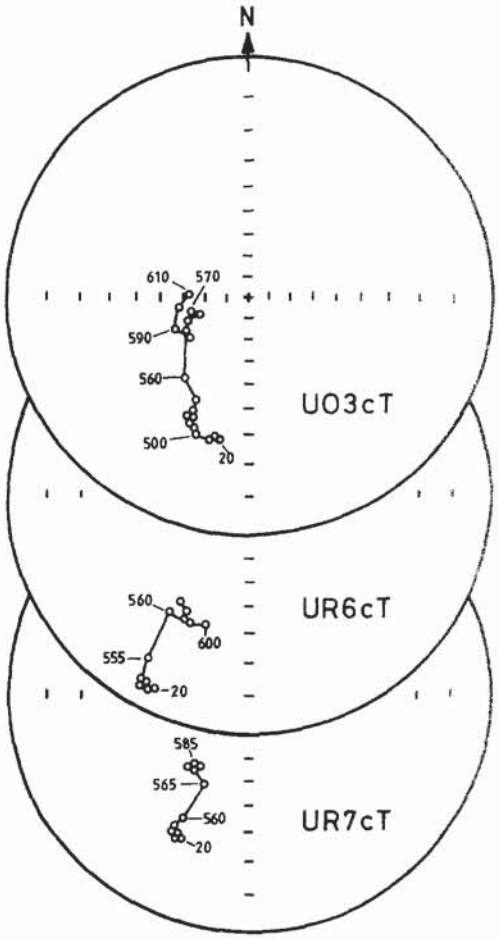


Fig. 6. Equal angle stereonets showing thermal demagnetization of Umvimeela Dyke specimens after the isolation of the S component by AF. The S component is progressively removed revealing, once again, the P component. All directions have negative inclinations.

4.4. Component Z

The NRMs of samples from the four northern sites within the deformed zone generally involve low magnetic moments (0.1×10^{-6} A m^2) and low initial susceptibilities. Most specimens were able to acquire a viscous remanent magnetization between sampling and measurements and had to be stored in Mumetal boxes. The specimens also tended to acquire rotational remanent magnetization in alternating fields less than 20 mT. The instability of the specimens made it difficult to determine the characteristics of NRM at these sites. Nevertheless, a distinct southerly downward component, which we label Z, was detected at sites UT and UU. At site UT an estimate of the Z direction was possible using demagnetization circle analysis and this gave $D=172.8°$, $I=62.9$ with $k=35.6$ and $\alpha_{95}=30.0°$ (Fig. 1). The corresponding virtual geomagnetic pole is at 41.8°E, 61.7°S.

4.5. Magnetic susceptibility

Site mean values for susceptibility are given in Table 1 and there are clearly two distinct groups. The first, involving seventeen of the sites, has

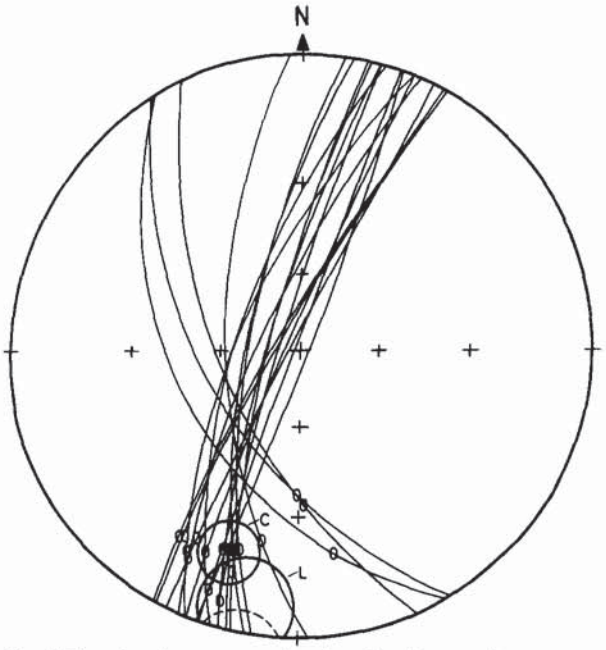


Fig. 7. Equal angle stereonet showing the nineteen demagnetization circles associated with the southerly swing from component P to component S. *C* is the 95% confidence circle about the mean direction determined by demagnetization circle analysis. For comparison, L shows the result determined by use of LINEFIND (see text).

values spread between 4.3×10^{-3} and 18.6×10^{-3} with a mean value of 8.2×10^{-3} (SI), while the four sites within the deformed zone have values closely grouped around 0.73×10^{-3}. At site UG the three samples from the sheared edge of the dyke have a mean of 1.8×10^{-3}, while the mean of the other five is 16×10^{-3}.

5. Discussion

5.1. Component P

Contacts of the Umvimeela Dyke with country rock were not located preventing the primary nature of P from being established, or otherwise, by a classic contact test. Nevertheless there is other evidence that suggests that P, carried predominantly by magnetite and present throughout the long cratonic section, is the primary component. Its direction is not significantly different from those of the Great Dyke, East Dyke and Popoteke gabbro. Thus if P is a later overprint, an event of almost regional extent is required. The Mashonaland Igneous Event (Wilson et al., 1987) was certainly of cratonic proportions and, during this event, a single site in the norite of the Great Dyke was completely remagnetized by a Mashonaland dolerite sill (McElhinny and Opdyke, 1964). Therefore, if P is not primary, it was introduced during a large-scale event occurring between the emplacement of the Great Dyke and the Mashonaland Igneous Event and there is no evidence of such an event. Finally comparisons of the blocking temperature spectra of P and those of thermoremanent magnetizations artificially introduced into the same specimens show virtually no differences (Mushayandebvu, 1991).

5.2. Component S

Unlike P there are noticeable variations of S both within-site and between-site. At site UG, S is more prominent in the sheared margin of the dyke than towards the undeformed middle. The magnetic susceptibility is an order of magnitude lower in the altered rock than at the dyke centre. S is absent at some sites but present in bigger proportions as the (younger) Zambezi and Limpopo Belts are approached. This suggests an origin related to activity in the Belts. The direction of S is unlike any known direction within the region older than 2500 Ma, but is similar to that of the 1100 Ma old Umkondo dolerites and lavas located principally 250 km to the east (there is an Umkondo dolerite dyke only 100 km to the east). S is probably a chemical remanent magnetization associated with Umkondo activity.

5.3. Component Z

Z was identified at two sites but its direction is well defined at only one of them. The corresponding virtual geomagnetic pole position, when compared to the apparent polar wander path for the Congo and Kalahari cratons of McWilliams and Kröner (1981) (Fig. 8), is roughly consistent with an age of ~650 Ma (i.e. the later part

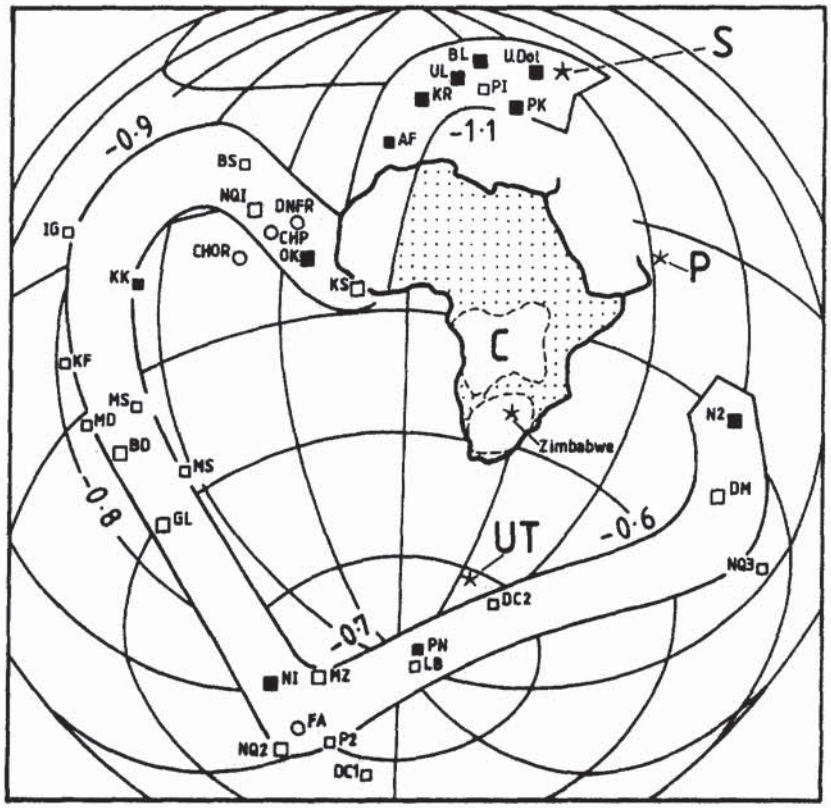


Fig. 8. A version of the Late Precambrian apparent polar wander path for the Kalahari and Congo cratons showing pole positions for the S and P components of the Umvimeela Dyke remanence and for site UT. The pole position for P coincides with that of the Great Dyke.

of Pan-African activity in the region; Mushayandebvu, 1991).

5.4. *Analysis of dispersion*

Using McFadden's (1982) technique, the P component of the Umvimeela Dyke has an angular dispersion of 6.5° with 95% confidence limits of 5.2° and 8.7°. This compares with a value of 9.8° (with 95% confidence limits of 7.0° and 16.1°) for the Great Dyke (Jones et al., 1975, recalculated by Mushayandebvu, 1991). This suggests that the time of emplacement and cooling of the Umvimeela Dyke was indeed less than that of the Great Dyke and was too short to average out secular variations as suggested by Jones et al. (1975). This idea is reinforced by the observation that the angular dispersion of the Umvimeela Dyke is significantly less than the angular dispersion of the present geomagnetic field (12.6° at latitude 39°; Brock, 1971), whereas the angular dispersion of the Great Dyke is not.

5.5. *Comparison of mean directions*

The mean direction for the P component at the seventeen sites in the Umvimeela Dyke is $D=219.8°$, $I=-58.7°$ with $k=116.8$ and $\alpha_{95}=3.3°$. That for the Great Dyke is $D=220.4°$, $I=-58.3°$ with $k=55.5$ and $\alpha_{95}=8.2$ (McElhinny and Gough, 1963). The mean directions for the two dykes are less than a degree apart and the circle of 95% confidence about the Umvimeela Dyke direction falls wholly within that about the Great Dyke mean. Clearly the two mean directions are indistinguishable. Mushayandebvu (1991) shows that the same conclusion

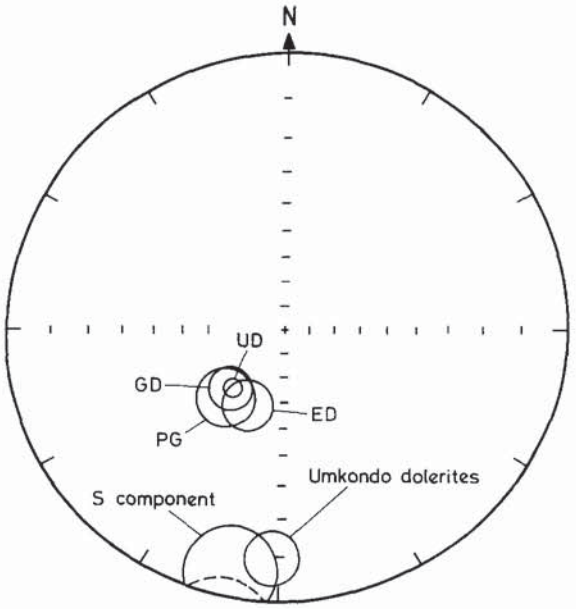


Fig. 9. Equal angle stereonet showing circles of 95% confidence about mean directions of various components and igneous bodies. *GD*=Great Dyke. *UD*=Umvimeela Dyke P component. *ED*=East Dyke. *PG*=Popoteke gabbro.

can be drawn for the East Dyke and Popoteke gabbro and the Great Dyke (Fig. 9), from comparisons of the mean directions.

The direction of the S component was estimated by two methods (see earlier) but it has not been completely isolated. The Umkondo dolerites, dated at 1080^{+140}_{-25} Ma (Allsopp et al., 1989), are thought to be part of regional-scale igneous activity and their direction of magnetization is $D=183°$, $I=-10°$ with $k=63$ and $\alpha_{95}=5.8°$. Fig. 9 shows the similarity in the directions of the S component of the Umvimeela Dyke and the characteristic remanence of the Umkondo dolerites. It is possible that the S component is associated with the Umkondo episode that preceded, or perhaps coincided with, the onset of the Pan-African activity during which the northern section of the Umvimeela Dyke and the Musengezi Complex were deformed.

6. Conclusions

Component P has been identified and defined at seventeen sites spread over the long cratonic section of the Umvimeela Dyke and at site US in the deformed zone at its northern end. The P direction is carried by single domain magnetite and, to a lesser extent, by ilmenohaematite. It is probably a primary thermoremanent magnetization. Analysis of the dispersion of site mean directions strongly suggests that this long dyke was emplaced and cooled in a short period of time.

Component S, carried by magnetite and, to a lesser extent, by titanomagnetite and ilmenohaematite, is more evident at the extremities of the Umvimeela Dyke where it enters the Zambezi and Limpopo Belts. Allsopp et al. (1989) suggested there had been widespread igneous activity throughout southern Africa 1100 Ma ago, termed the Umkondo Igneous Event. The similarity of S and the Umkondo dolerite direction is consistent with this hypothesis. The event probably included the flushing of fluids through many of the rocks of the sub-continent resulting in chemical alterations and the acquisition of the S component. More thorough studies of pre-Mesoproterozoic formations in southern Africa may reveal a widespread occurrence of component S and hence help define the geographic extent of the Umkondo Event.

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